MIL-H-29144B
8 April 1985
SUPERSEDING
MIL-H-29144A
30 June 1977

MILITARY SPECIFICATION

HOOD, FIREMEN'S, ALUMINIZED, PROXIMITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a heat protective hood constructed from an aramid outer fabric having a highly reflective aluminized face.

1.2 Classification. The hood covered by this specification shall be of one size only and of following types as specified (see 6.2):

Type I - Hood, Shoulder Length (Aircraft Firefighting)
Type II - Hood, with Front Apron (Shipboard Damage Control)

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

* SPECIFICATIONS

FEDERAL

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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L-P-377	- Plastic Sheet and Strip, Polyester
FF-R-556	- Rivets, Solid, Small; Rivets, Split, Small; Rivets Tubular, Small; and Caps, Rivets; General Purpose
FF-W-92	- Washers, Metal, Flat (Plain)
JJ-W-155	- Webbing, Textile (Cotton, Elastic)
QQ-N-290	- Nickel Plating (Electrodeposited)
QQ-P-416	- Plating, Cadmium (Electrodeposited)
CCC-C-440	- Cloth, Cheesecloth, Cotton, Bleached and Unbleached
DDD-L-20	- Label; For CLothing, Equipage, and Tentage (General Use)
GGG-H-142	- Helmet; Construction Worker's
PPP-B-636	- Boxes, Shipping Fiberboard
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain for Sealing and Securing

MILITARY

MIL-W-530	- Webbing, Textile, Cotton, General Purpose, Natural or in Colors
MIL-B-543	Buckles, Tongueless and Web Strap
MIL-F-1148	- Fibre, Vulcanized, Electrical and Mechanical Grades
MIL-R-3390	- Rings, Dee
MIL-P-8184	- Plastic Sheet, Acrylic, Modified
MIL-C-19699	- Cloth, Coated (Nylon Taffeta)
MIL-T-43566	- Tape, Textile, Cotton, General Purpose, Natural or in Colors
MIL-T-43636	- Thread, Aramid
MIL-T-44100	- Thread, Para-Aramid, Spun, Intermediate Modulus
MIL-B-87002	- Batting, Synthetic Fibers, Quilted Aramid
MIL-C-87076	- Cloth, Coated, Aramid, Aluminized

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-1666	- Classification of Defects for Evaluation of Quality of Aluminized Firemen's Clothing
MS-27980	- Fasteners, Snap, Style 2

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING MATERIALS
ASTM-D-1003-61 - Haze and Luminous Transmittance
of Transparent Plastic, Test for

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

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* 3.3.1 Basic material. The basic material for the outer shell of the hood shall be an aramid material coated with a vacuum deposited aluminized face conforming to MIL-C-87076.

* 3.3.2 Quilted lining. The material for the quilted lining shall be aramid quilted batting conforming to type I of MIL-B-87002 except the use of recycled material (batting) is prohibited. The outer covering cloth conforming to MIL-C-43774 shall have 69 filling yarns and the dimensional stability for warp and filling shall be 3 percent maximum.

* 3.3.3 Thread. The thread for assembling the outershell and for joining the outershell to the linings shall be non-melting nylon (Aramid), natural shade, conforming to Size B and E of MIL-T-43636. As an alternate, a thread conforming to size T-50 of MIL-T-44100, natural in color may be used.

3.3.4 Interlining. The material for interlining the quilted lining of the hood shall be coated nylon taffeta conforming to type II of MIL-C-19699 except the resistance to crocking shall be not less than "fair".

3.3.5 Elastic take-up strap (Type I Hood). The material for the back take-up strap shall be elastic cotton webbing conforming to type I, class 5, 7/8 inch wide of JJ-W-155 except that the colorfastness requirements shall not apply.

3.3.6 Snap fasteners. The snap fasteners for attaching the hood to the frame and closing the side flaps to the face opening shall consist of two separate components (button cap and socket) that constitute the female half of the fastener and two separate components (stud and eyelet) that constitute the male half of the fastener. The fasteners shall be style 2, 24-line, size 1, button conforming to MS27980 - 1N or 1C; socket conforming to MS 27980 - 6N or 6C; stud conforming to MS 27980 - 7N or 7C; and eyelet conforming to MS 27980-8N or 8C.

3.3.7 Chin and harness (take-up) adjustment strap assembly.

3.3.7.1 Chin (take-up) adjustment strap assembly. The chin strap assembly shall be as shown in Figure 6 and shall consist of a tongueless buckle and cotton tape.

3.3.7.1.1 Buckle. The buckle shall be tongueless, double bar with lip, conforming to type II, style 3, class 1 or 3, 5/8 inch size of MIL-B-543. The class 3 buckle shall have a nickel or chrome finish.

3.3.7.1.2 Tape. The tape shall be cotton, natural, bleached or dyed shade Olive Drab conforming to type I, classes 1, 2, or 3, 5/8 inch wide of MIL-T-43566, except that the colorfastness requirements shall not apply.

3.3.7.2 Harness (take-up) strap assembly (Type II Hood). The harness strap assembly shall be as shown in Figure 2 and shall consist of a tongueless single bar buckle, webbing, dee rings, and a snap secured by rivets.

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3.3.7.2.1 Buckle. The buckle shall be tongueless, single bar, conforming to type I, style 2, class 1, 3/4 inch size, construction A or B of MIL-B-543, except that the finish for the brass shall be nickel plate conforming to class 2, 0.0005 inch minimum thickness of QQ-N-290.

3.3.7.2.2 Webbing. The webbing shall be cotton, natural, bleached or dyed Olive Drab, type IIa, classes 1, 2 or 3, 3/4 inch wide conforming to MIL-W-530, except that the colorfastness requirements shall not apply and it shall have an approved durable flame retardant treatment (see 4.4.1 and 6.4). As an alternate, a high temperature resistant aromatic-polyamide (aramid) webbing may be used. A 3/4 inch 100% filament yarn (aramid) webbing, identified as Pattern 1174, may be obtained from Bally Ribbon Mills, Bally, PA (See 4.4.1).

3.3.7.2.3 Rings, dee. The dee rings shall be class 1 or class 3, style I, symbol X, 3/4 inch wide conforming to MIL-R-3390. The finish for the malleable iron shall be cadmium plate conforming to class 3, type I of QQ-P-416. The finish for the brass shall be nickel plate conforming to class 2, 0.0005 inch minimum thickness of QQ-N-290.

3.3.7.2.4 Snap. The snap for fastening the harness assembly shall be cast malleable iron, brass or cast aluminum conforming to the material and dimensional requirements specified in Figure 7 (see 4.4.1). The finish for the malleable iron shall be cadmium plate conforming to class 3, type I of QQ-P-416. The finish for the brass shall be nickel plate conforming to class 2, 0.0005 inch minimum thickness of QQ-N-290.

3.3.7.2.5 Rivets, tubular with caps or washers. The rivets for attaching the snap shall be 9/64 inch nominal diameter with 5/16 inch nominal length, conforming to type XII, class 3, grade B or F of FF-R-566, except that they shall be drilled to the head or partially drilled to the head and the inside of the rivet shank shall be chamfered. The caps shall conform to type II of FF-R-556. The finish for the brass rivets and caps shall be nickel plate conforming to class 2, 0.0005 inch thickness of QQ-N-290. An aluminum washer conforming to type A, grade II, class C, No. 8 of FF-W-92 may be used in lieu of caps for securing the tubular rivets.

3.3.8 Vulcanized fiber. The material for stiffening the aluminized outer facepieces and side flaps shall be 1/16 inch thick and for the chin rest shall be 3/32 inch thick vulcanized fiber conforming to grade BH or CH of MIL-F-1148. As an alternate, the chin rest may be made of other rigid and lightweight materials that meet the performance requirements of MIL-F-1148. The chin rest shall be molded to conform to the dimensions shown in Figure 5. The chin rest shall be connected to the frame in such a manner that it may be adjusted to the desired position in front of the face. The chin rest shall be located at the drill marks shown in Figure 4. The chin rest shall be held firmly in place after being adjusted.

3.3.9 Hood facepiece assembly.

3.3.9.1 Facepiece support. The material for the facepiece support shall be methyl methacrylate, 0.125 (± 0.015) inch thick conforming to finish A of MIL-P-8184. The facepiece support shall be formed and sized to the shape and dimensions shown in Figure 8.

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3.3.9.2 Metallized facepiece. The material for the facepiece shall be polyester type plastic film conforming to type III of L-P-377, except that the thickness shall be 0.007 (± 0.001) inch and the maximum haze factor shall be 9.2 percent. The facepiece shall be gold coated on one surface and then overcoated with an abrasion resistant coating (see 4.4.1) by vacuum deposition or other suitable techniques. The gold coated side shall be identified with the word "outside" printed in black indelible ink letters of not less than 1/4 inch high along the edge of the sheet so that it is covered by the outer facepiece facing when in place. The facepiece shall measure 14 (± 1/8) by 8-1/4 (± 1/8) inches and shall be positioned onto the hood by inserting it over the facepiece support so that the "outside" surface is away from the facepiece support (see Figure 3).

3.3.9.3 Performance. The performance of the facepiece and facepiece support shall be as indicated below when tested as specified in 4.4.1.

3.3.9.3.1 Vertex power and prismatic power. The vertex power of the facepiece support, excluding the area 3/4 inch from the outer edge, shall not exceed the inherent power to the required curvature and thickness by more than ± 0.06 diopters in any horizontal meridian when tested as specified in 4.4.1.1. The inherent power of the facepiece support shall be calculated by use of the following equation:

$$F = (F_1 + F_2) \frac{-t}{n} F_1 \times F_2)$$

$$F_1 = \frac{n-1}{r_1}, \quad F_2 = \frac{1-n}{r_2}$$

or

$$F = \frac{-t}{r_2 (r_2 + t)} \left[1 - \frac{1}{n}\right]$$

F = Inherent power of the facepiece support in diopters.

F_1 = Power of the front surface of the facepiece support in diopters.

F_2 = Power of the back surface of the facepiece support in diopters.

n = Index of refraction of the facepiece support material.

t = Thickness of facepiece support in meters.

r_1 = Radius of the front surface of the facepiece support in meters.

r_2 = Radius of the back surface of the facepiece support in meters.

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The vertical prismatic components at any two points on any given horizontal meridian of the facepiece support excluding the area 3/4 inch from the outer edge shall not differ algebraically by more than 0.125 prism diopter and the absolute vertical prismatic deviation measured from any point in any horizontal meridian shall not exceed 0.125 prism diopter when tested as specified in 4.4.1.1.

3.3.9.3.2 Optical distortion. The facepiece and the facepiece support shall be free from any visible distortion or optical defects when tested as specified in 4.4.1.2.

3.3.9.3.3 Luminous transmittance. The luminous transmittance of the facepiece shall be not less than 20 percent (illuminant C) when tested as specified in 4.4.1.3.

3.3.9.3.4 Adhesion of coatings. The coatings on the facepiece shall not be removed when subjected to the adhesion test specified in 4.4.1.4.1. If there is visual evidence that some of the coatings were removed, the tested area of the facepiece shall be tested again for optical distortion, hardness of coatings, and infrared radiant heat resistance. The facepiece shall be acceptable if it meets the requirements of the above tests after the adhesion test.

3.3.9.3.5 Hardness of coatings. The coatings on the facepiece shall not be removed or show visible evidence of marring or abrasion when tested as specified in 4.4.1.4.2. If there is a visual evidence that some marring or coating removal has occurred, the tested area of the facepiece shall be tested again for optical distortion, luminous transmittance, and infrared radiant heat resistance. The facepiece shall be acceptable if it meets the requirements of the above tests after the hardness test.

3.3.9.3.5.1 Cloth cheesecloth. The cheesecloth used in testing the hardness of coating for the metallized facepiece shall conform to type I, class 2 of CCC-C-440 (see 4.4.1.4.2).

3.3.9.3.6 Infrared transmittance. The infrared transmittance of the facepiece shall be no greater than 10.0 percent between 1.0 and 7.5 microns when tested as specified in 4.4.1.5.

3.3.9.3.7 Infrared radiant heat resistance. The gold coated film shall show no holes or visual evidence of cloudiness or dullness when tested as specified in 4.4.1.6.

3.3.9.4 Facepiece cover.

3.3.9.4.1 Cover. The facepiece cover shall be made of a commercial grade of flexible plastic film of 0.020 inch thickness plus or minus normal commercial thickness tolerances. The color of the film shall be dark and opaque. The cover shall be positioned on the hood with six 5/8 inch diameter attachment holes placed over the corresponding snap fastener studs on the aluminized facepiece. The finished facepiece cover shall measure 14-3/8 by 8-1/2 (± 1/8) inches.

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3.3.9.4.2 Marking. The printed marking on the outside surface of each facepiece cover shall be bright yellow or orange color conforming to type IV, class 9 of DDD-L-20. The marking shall be legible, have good color retention with no cracking or peeling after 4 hours water immersion (see 4.4.1.7). The ink shall show good resistance to water, wet and dry crocking (see 4.4.1). The marking shall be centered on the cover in block letters of not less than 3/4 inch high and 1/2 inch wide and shall contain the following information (see Figure 3):

REMOVE BEFORE USE. REPLACE AFTER USE.

3.3.10 Identification and instruction label. Each hood and cover (as applicable) shall have a combination identification and instruction label conforming to type VI, class 15, of DDD-L-20, except that the colorfastness requirements shall not apply. The label shall bear the following inscription:

COVER, FIREMEN'S HOOD FRAME REPLACEABLE FOR
HOOD, FIREMEN'S, ALUMINIZED, PROXIMITY
TYPE I (EXAMPLE)
BASE FABRIC: ARAMID (100%)
CONTRACT NO.: DLA-100-00-0-0000 (EXAMPLE)
STOCK NO.: 8415-00-000-0000 (EXAMPLE)
NAME OF CONTRACTOR:

USE AND CARE INSTRUCTIONS

PURPOSE

The gold-coated facepiece and aluminized materials are reflective heat shields. The facepiece is NOT a sun shield. Both items must be kept in good condition to reflect the heat radiation efficiently. In particular, when the facepiece's gold surface becomes worn, scratched or marred, it should be immediately replaced, because 90% of the heat protection may be lost.

CARE

Facepiece

1. Keep protective facepiece cover in place when carrying or storing the hood to minimize damage to gold-coated surface. Remove when using hood.

2. For adequate protection, replace worn out gold-coated facepiece. Make sure the gold surface is on OUTSIDE as marked on edge.

3. Avoid touching or wiping the gold surface as much as possible.

4. After each use, clean the facepiece, without removing it from the hood, using a clean, soft cloth with mild soapy water, rinse, and pat dry.

Aluminum Material

1. Hang or store hood properly to prevent damage to aluminized surface. Clean by gently rubbing aluminized surface with a soft cloth using either a mild soapy water, ISOPROPANOL or PERCHLORETHYLENE.

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2. Do not use aluminized paint to repair worn surface as it will not provide protection.

NOTE: Aluminized coating is damaged by contact with Potassium Bicarbonate Powder (PKP). Before storing garment with PKP residue present, the following is recommended:

1. Dry powder should be removed with a clean soft brush or blower.

2. Wet powder should be removed by thoroughly flushing with water and wiping the surface down with a clean soft cloth. Damp or wet hoods should be hung to dry.

REPLACEABLE STOCK HOOD COMPONENTS

1. Facepiece, Firemen's Hood (Gold Coated); NSN 8415-00-001-6489
2. Facepiece Support, Firemen's Hood (Clear Plastic); NSN 8415-00-782-2115
3. Cover, Firemen's Hood, Frame (Aluminized); NSN 8415-01-131-7913

SOURCE - Defense Logistics Agency, 2800 South 20th Street, Philadelphia, PA 19101

NOTE: Labels on Type II hoods shall not contain reference to the third replaceable component.

3.3.11 Hood support assembly.

3.3.11.1 Frame. The frame shall be made of fibrous glass reinforced polyester resin. The frame shall be free from uncured or unbonded areas, voids, cracks, holes, blisters, excess surface resin, air or gas pockets, and other similar defects. The frame shall possess sufficient strength to support the attachment of the helmet and hood and have sufficient stiffness to maintain its curvature when the hood is fastened to the frame. The frame shall be a one-piece molded type and shall conform to the dimensions and thickness shown in Figure 4 (see 4.4.1).

3.3.11.2 Helmet. The helmet shall conform to type II of GGG-H-142 and shall be adjustable from sizes 6-3/4 to 7-3/4 inclusive. Insulation resistance is not required. The helmet shall be readily attachable to the frame by means of a swivel type spring-loaded plunger attachment (plunger, cup and plug assembly) conforming to "QUICK LOK No. 4000," manufactured by Fibre-Metal Product Company, Concordville, PA 19331 or equal (see 6.5).

3.3.11.3 Function of assembly. The hood support assembly shall function as follows:

a. The assembled frame shall project beyond the face to permit clearance for nose, eyeglasses (when used), and full face oxygen breathing apparatus (when used).

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b. The assembled frame without hood can be easily tilted upward to a point where the center of the bottom edge of the frame will be out of the line of vision and projected from the eyes at an angle of at least 30 degrees above horizontal and shall hold firmly in place after being adjusted.

c. The tightness of the spring-loaded plunger attachment shall be adjustable.

d. All attachments and adjustment fittings shall be corrosion resistant having sufficient strength for the purpose intended. They shall be capable of adjustment by hand, without the use of tools. These fittings shall be of the best quality suitable for the purpose intended.

* 3.3.12 Repairing aluminized surface. The unsupported aluminum film used for repairing uncoated or cracked areas on the finished hood shall be the same as specified for coating the aramid base cloth. The area to be repaired shall not exceed 1 inch in the longest direction. The repair patch shall be completely bonded onto the aluminum film of the hood and shall be so placed that it extends not less than 1/2 inch beyond all edges of the area to be repaired. No hood shall contain more than 6 patches. The unsupported aluminum film and self-curing adhesive used for repairing shall be obtained from the basic material supplier (see 3.3.1). The size of the patch shall not exceed 2 inches in any direction.

* 3.4 Design. The hood, including inserts, frame, chin rest, and the attachment of the chin and harness (take-up) adjustment strap assemblies, hanger loop, and a covered elastic take-up strap shall be as shown in Figures 1 through 8. The hood and cover shall consist of an outer shell fabricated from an aramid base cloth having a highly reflective aluminized outer surface. The outer shell shall be joined to a quilted lining inner shell which provides thermal insulation and waterproofness. The hood and cover shall have a channeled face opening to permit easy insertion and removal of the facepiece support and metalized facepiece. The inserts are securely held in position by two flaps, one at each end of the face opening that are fastened down by three snap fasteners. Also held in place by the flaps is a facepiece protective cover. The aluminized hood shall be attached to an inner frame (with helmet) by means of six snap fastners. Type II hood shall have an apron to cover and protect an oxygen breathing apparatus.

3.4.1 Figures. The figures are considered a part of this specification. When inconsistencies exist between the written specification and the figures, the written specification shall control.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The patterns allow a 3/8 inch seam (aluminized outer shell assembly) and 1/2 inch seam (quilted lining-interlining inner shell assembly), except as otherwise indicated in Table I. The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way.

3.5.1 Pattern parts. The component parts for the hoods shall be cut from the materials specified in accordance with the following pattern parts:

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Material	Nomenclature of Pattern Parts	Cut Parts
	(TYPE I HOOD)	
Cloth, coated, aramid, aluminized (outer shell)	Hood	1
	Outer facepiece	1
	Outer facepiece facing	1
	Flaps (side)	2
	Facing, bottom 1/	1
	Elastic take-up strap	1
	Hanger loop	1
	Facing, inside facepiece	1
Cloth, coated, nylon taffeta (interlining)	Hood	1
Quilted assembly (lining)	Hood	1
Vulcanized fiber	Outer facepiece (stiffner)	1
	Side flaps (stiffner)	2
	(TYPE II HOOD)	
Cloth, coated, aramid, aluminized (outer shell)	Hood 2/	1
	Outer facepiece	1
	Outer facepiece facing	1
	Flaps (side)	2
	Facing, back	2
	Facing, side 1/	2
	Facing, bottom 1/	1
	Hanger loop	1
	Facing, inside facepiece	1
Cloth, coated, nylon taffeta (interlining)	Hood 2/	1
Quilted assembly (lining)	Hood 2/	1
Vulcanized fiber	Outer facepiece (stiffener)	1
	Side flaps (stiffener)	2
Template	Location marker for snap hood and underarm adjustment strap	

1/ Pattern may be cut as two pieces with a 3/8 inch seam allowance added for each piece.

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2/ When material width limits the hood from being cut from one piece of material, the bottom of the bib may be pieced with seam allowance added. When pieced, the maximum finished width of the piece shall be 6 inches, measured from the bottom finish edge of the bib. There shall be no more than one piecing per hood. Standard specification joining seam construction shall be used.

3.6 Construction.

3.6.1 Stitch, seam and stitching types. Stitch, seam, and stitching types specified in Table I shall conform to FED-STD-751. Wherever two or more methods for seam or stitch types are given for the same part of the operation, any one may be used. Where stitch type 401 is used, the ends of stitching shall be caught in other seams or row(s) of stitching and the looper (underthread) shall be on the inside of the hood. Topstitching and edgestitching shall be 1/8 to 3/16 inch from the turned edge, except edgestitching of dome seam which shall be 3/16 to 1/4 inch from the turned edge. In order to minimize the size of needle hole in aluminized fabric, the needle used in the seaming operation shall be of the smallest size which will accommodate the specified size of thread. Wherever LSq-2 stitch type is specified, the top ply shall not be turned back sharply over the first row of stitching but turned in such a manner that a loose welt is formed. The bight of the serge stitching shall be 3/16 to 1/4 inch.

3.6.2 Thread breaks and ends of seams. Ends of all seams and stitchings when not caught in other seams or stitchings shall be backstitched not less than 3/8 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks (301 or 401 stitch types) may be repaired by 301 stitch type with the stitching anchoring both ends of the break.

3.6.3 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table I.

3.7 Manufacturing operations requirements. The hoods shall be manufactured in accordance with the operation requirements specified in Table I. The contractor is not required to follow the exact sequence of operations listed provided that the finished hoods are identical to those produced by following the sequence of operations as listed in Table I.

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3.7.1 Abbreviations in table of operations. The abbreviations used in table I are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper
Mchne	-	Machine
Brtck	-	Bartack
Comrcl	-	Commercial
smlr	-	similar
Btnhl	-	Buttonhole
incl	-	including
dbl	-	double

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TABLE 1 - CONSTRUCTION OF HOODS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
1.	Cutting				
	a. The hood shall be cut in strict accordance with patterns furnished which show placement of component parts, directional lines for cutting, and notches for proper assembly of all parts. A maximum swing of 2-1/2 inches from the directional line is permitted. Measurements shall be taken at the top and bottom of the directional line on the pattern to the selvage edge of the fabric on the even side of the lay. The difference between the two measurements shall not exceed 2-1/2 inches. Drill holes are permitted for positioning provided they are covered in the finished hood. NOTE: Patterns allow the quilted batting to be cut in any direction.				
	b. The component parts of the aramid cloth may be cut from ends and when so cut shall finish with the aluminized side face up. A single splice on the side and bottom facings is permitted.				
	c. The quilted lining shall be cut in such a manner so that the aramid cloth side is against the uncoated side of the nylon taffeta interlining.				
	d. The coated nylon taffeta interlining shall be cut in such a manner so that the coated side of the material is against the uncoated side of the outer shell.				
	e. Cut elastic webbing for back take-up strap not less than 13-1/2 inches in length (Type I Hood).				
	f. The face side of the materials to be assembled shall be considered as follows:				
	1. Outershell - aluminized side				
	2. Quilted batting - black rayon side				
	3. Interlining - coated side				

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TABLE 1 - CONSTRUCTION OF HOODS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
2.	Replacement of defective components During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-1666 shall be removed from production and replaced with nondefective and properly matched components.					
3.	Prepare aluminized outer shell assembly Dome seam joining the left and right sections of the hood shall be uniformly stitched and carefully sewn to avoid twisting or pleating and finished evenly. Attached component parts shall be properly positioned and the shape of the face opening shall be as shown by marks on patterns.					
	a. Position, face to face, the inside facepiece facing along the face opening of the hood as indicated by marks on pattern. Stitch along all four sides on the outside drill marks.	301	LSq-2(a)	6-8	B or E	B or E
	b. Cut both plies to material on the inside drill marks and slash the seam allowance diagonally at the four corners.					
	c. Turn and edgestitch 1/8 to 3/16 inch from the turned edge.	301	LSq-2(b)	6-8	B or E	B or E
	d. Join face to face, 3/8 (+ 1/16 inch) from the raw edge the dome seam of the right and left sections of hood. Turn joining seam face side up to form a loose welt and stitch 3/16 to 1/4 inch from the turned edge. An adjustable hinged raising foot may be used in forming the loose welt seam. The right section shall overlap the left section. NOTE: Slitting and spreading of lapped seam for a distance of approximately 1 inch shall be permitted at the dome and dome dart seam junction.	301	LSq-2	6-8	B or E	B or E

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TABLE 1 - CONSTRUCTION OF HOODS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
	e. Join face to face, 1/2 (± 1/16) inch from the raw edge the horizontal dome darts of the right and left sections of the hood as indicated by notches on pattern in one continuous operation. Turn joining seam and edgestitch 1/8 to 3/16 inch from the turned edge with the fold towards the face opening.	301	LSq-2	6-8	B or E	B or E
	f. Join, 1/2 (± 1/16) inch from the raw edge, the vertical dome darts of the right and left sections of the Type II hood as indicated by notches on pattern. Turn joining seam and edgestitch 1/8 to 3/16 inch from the turned edge with the fold towards the face opening.	301 or 401	LSq-2	6-8	B or E	B or E
4.	Prepare inner shell assembly The uncoated side of the nylon taffeta interlining shall be against the aramid cloth side of the quilted lining. The dome darts shall be carefully tapered to avoid unsightly bulging and ends securely tacked					
	a. Join the quilted lining and coated interlining together by stitching 3/16 to 1/4 inch from the raw edge or by overedge stitching all raw edges of assembly.	301 or 401 or	SSa-1	6-8	B or E	B or E
		502 or 503 or 504	SSa-1	6-10	B or E	B or E
	b. Cut out each ply of quilted lining and coated interlining material on the drill marks.					
	c. Join face to face, 1/2 (± 1/16) inch from the raw edge, the horizontal dome darts of the right and left section of the hood lining as indicated by notches on pattern in one continuous operation.	301 or 401	SSa-1	6-8	B or E	B or E

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TABLE 1 - CONSTRUCTION OF HOODS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
	d. Join face to face, 1/2 (± 1/16) inch from the raw edge, the vertical dome darts of the right and left sections of the hood lining as indicated by notches on pattern (Type II hood).	301 or 401	SSa-1	6-8	B or E	B or E
	e. Join face to face, 3/8 (± 1/16) inch from the raw edge, the back and side aluminized facing pieces. Spread seam and stitch 1/8 to 3/16 inch from each side of the joining seam (Type II hood).	301 or 401	SSz-3	6-8	B or E	B or E
	f. Join, face to face, 3/8 (± 1/16) inch from the raw edge, the side and bottom aluminized facing pieces, terminating the stitching approximately 1/2 inch to allow for turning under. Turn bottom aluminized facing piece to finished position (Type II hood).	301 or 401	SSa-1	6-8	B or E	B or E
	g. Position the bottom of aluminized facing pieces, as applicable, on the rayon side of the quilted assembly and stitch together along outside raw edges 1/8 to 3/16 inch from edge. NOTE: Spliced facing shall be prestitched.	301 or 401	SSa-1	6-8	B or E	B or E
	h. Turn under 3/8 to 1/2 inch the inner edge of aluminized facings and topstitch 1/8 to 3/16 inch from the folded edge to quilted lining assembly.	301 or 401	LSd-1	6-8	B or E	B or E
	i. Attach label:					
	1. On Type I hood, simultaneously catch bottom edge of label while performing topstitching under operation 4.h. The label shall be positioned at back of hood on either the left or right section approximately 1/2 to 1 inch from dome seam. Stitch 1/16 to 1/8 inch from the edge the remaining three sides of label.	301	LSbj-1	6-8	B or E	B or E

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TABLE 1 - CONSTRUCTION OF HOODS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
	2. On Type II hood place label on quilted lining at the front and center of hood with bottom edge of label approximately 1 to 2 inches above folded edge of aluminized facing. Stitch all four sides 1/16 to 1/8 inch from edge. The bottom edge of label may be simultaneously caught when stitching the turned inner edge of aluminized facing.	301	LSbj-1	6-8	B or E	B or E
	j. Join, face to face, 1/2 (± 1/16) inch from the raw edge, the dome seam of the right and left sections of the hood, catching the free end of aluminized facing between the plies.	301 or 401	SSa-1	6-8	B or E	B or E
	k. Spread aluminized facing seam open and stitch 1/8 to 3/16 inch from each side of the joining seam	301 or 401	SSz-3	6-8	B or E	B or E
	OR					
	1. As an alternate, the facing seams may be turned to one side and stitched 1/8 to 3/16 inch from turned edge. NOTE: When the bottom facings are cut in two pieces the joining shall follow operations k or 1.	301 or 401	LSq2	6-8	B or E	B or E
*5.	Join aluminized and lining-interlining hood assemblies					
	a. Insert lining-interlining assembly into assembled outer shell. Position inside facepiece facing to the underside over the quilted lining at the face opening. Turn in the free edges 3/8 to 1/2 inch and topstitch 1/8 to 3/16 inch from the turned edge along all four sides through all plies of material.	301	LSd-1	6-8	B or E	B or E

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TABLE 1 - CONSTRUCTION OF HOODS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
	b. Turn outside edge of aluminized shell over the aluminized facing prejoined to the lining-interlining assembly. Turn in raw edge 3/8 to 1/2 inch and stitch 1/8 to 3/16 inch from the turned edge forming a 3/4 to 7/8 inch hem along the periphery.	301	LSd-1	6-8	B or E	B or E
	c. The finished apron for Type II hood shall measure 25 (+ 1) inches from the top edge of the bottom panel of the face opening to the bottom hemmed edge. NOTE: When width of aluminized material is 40 inches or less, an additional piece of material shall be sewn to the bottom of apron to meet the finished 25 inch requirement. The additional piece shall finish not less than 2 inches nor more than 3-1/2 inches in width when measured from seamline to bottom hemmed edge of apron. The joining of the piece shall follow operations 4.k. or 4.l.					
	d. Attach six button caps and sockets as indicated by marks on pattern with the button caps on the exposed side of the inner face opening.					
6.	Make and attach hanger loop					
	a. Fold hanger loop part in accordance with marks on pattern. Turn under top layer 1/4 to 5/16 inch along its length and stitch 1/8 to 3/16 inch from folded edge down center of loop.	301	Similar to EFj-1	6-8	B or E	B or E
	b. Position and center the formed hanger loop horizontally across the center back seam of facing with bottom edge of loop 1-1/2 to 1-3/4 inches from folded bottom edge of hood. Turn under ends as indicated by marks on pattern and stitch to hood through all plies with a 5/8 (+ 1/16) inch cross boxstitch.	301	Box-criss-cross	6-8	B or E	B or E
	c. The finished hanger loop shall measure 5-1/4 (+ 1/8) inches long and 3/4 to 7/8 inch wide.					

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TABLE 1 - CONSTRUCTION OF HOODS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
7.	Make and attach outer aluminized facepieces					
	a. Position the outer aluminized face opening pieces, face to face, as indicated by marks on pattern with the facepiece facing superimposed on the facepiece. Stitch along all four sides of opening on the outside drill marks.	301 or 401	LSq-2(a)	6-8	B or E	B or E
	b. Cut out both plies of aluminized material on the inside drill marks and slash the seam allowance diagonally at the four corners.					
	c. Turn and edgestitch 1/8 to 3/16 inch from the turned edge.	301	LSq-2(b)	6-8	B or E	B or E
	d. Insert the vulcanized fiber between the two plies of aluminized material. Turn in outside aluminized face opening pieces at right and left sides. Stitch periphery of the vulcanized fiber with stitching 1/8 to 3/16 inch from the folded side edges and just off of the vulcanized fiber on the top and bottom edges. The vulcanized fiber shall not be caught in the stitching.	301	SSv-1 and SSc-1	6-8	B or E	B or E
	e. Attach three studs and eyelets of snap fasteners on the right and left ends of the assembled face opening piece as indicated by marks on pattern with studs on the outside.					
	f. Turn under 7/16 to 9/16 inch the top and bottom edges of outer facepiece and stay-stitch 1/8 to 3/16 inch from the folded edge.	301	EFa-1	6-8	B or E	B or E
	g. Position outer facepiece on hood with inside and outside face openings aligned with each other. Bottom and top folded edge shall be superimposed on inner face piece stitching. Stitch top and bottom edges of outer facepiece to hood along stitchline of staystitched edge.	301	LSd-1	6-8	E	E

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TABLE 1 - CONSTRUCTION OF HOODS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
8.	Make and attach aluminized side flaps					
	a. Fold each aluminized side flap lengthwise, face to face, as indicated by notches on pattern and stitch ends 1/2 inch from the raw edge.	301	SSa-1	6-8	B or E	B or E
	b. Turn to finished position and insert the side flap vulcanized fiber. Stitch entire length of each flap adjacent to vulcanized fiber insert.	301	LSbj-1	6-8	B or E	B or E
	c. Position wider portion of flap face down on hood with raw edge of seam extension facing towards front of hood, parallel to and 1-3/4 (± 1/16) inch from side of face opening. Stitch length of flap to hood 1/8 to 1/4 inch from raw edge. Fold flap to finished (closed) position and topstitch length of flap to hood, simultaneously covering raw edge of flap. The line of topstitching shall be superimposed on inner face piece stitching so that when fully fastened each flap shall have a vertical elevation of 3/4 to 7/8 inch along its hinge portion.	301	LSs-2	6-8	B or E	B or E
	d. Attach three buttons and sockets to each side flap as indicated by marks on pattern with the buttons on the outside corresponding to studs on the right and left ends of assembled face opening piece.					
	e. The finished flaps shall measure 9 (± 1/8) inches long and 2 (± 1/8 inches wide).					

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TABLE 1 - CONSTRUCTION OF HOODS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
9.	Make and attach rear tape-up strap (Type I Hood)	301	SSa-1	6-8	B or E	B or E
	a. Position a single piece of elastic webbing as indicated by notches and marks on pattern, on the uncoated side of the rear take-up strap 1/4 inch below center fold and on the underply side (as formed). Stitch ends 1/4 inch and 2 inches from edge. Fold rear take-up strap in half lengthwise with the aluminized face side in. Stitch the ends 1/2 (+ 1/16) inch from the raw edge simultaneously catching the elastic webbing and terminating the seam 1/2 inch from the raw edge.					
	b. Trim corners and turn to finished position working out corners. Turn in bottom raw edges 1/2 inch and stitch 1/8 to 3/16 inch from the turned edge. The elastic webbing shall not be caught in the seaming.	301	SSc-1	6-8	B or E	B or E
	c. Stitch the rear take-up strap to hood as indicated by marks on patterns, forming a 1-7/16 (+ 1/16) inch wide by 2-3/4 (+ 1/4) inches long cross boxstitch.	301	Box-criss-cross stitch	6-8	B or E	B or E

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TABLE 1 - CONSTRUCTION OF HOODS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
10.	Make and attach chin (take-up) strap assembly					
	a. Thread, fold, stitch and attach chin (take-up) adjustment strap assembly to each inner side of helmet with a suitable type of metal rivet positioned approximately 1/2 inch below and 1 inch from the plug portion of the plunger attachment toward the brimless area as indicated by figure 6.	301	SSa-1 and EFb-1 Box-criss-cross stitch	10-14	B or E	B or E
	b. The finished length shall be as follows: Long strap - 18 (± 1/2) inches Short strap - 2-1/8 (± 1/8) inches Long strap shall be measured from outer folded edge of hem to fold at inner left side of helmet (as worn). Short strap shall be measured from fold over buckle to fold at inner right side of helmet (as worn).					
11.	Make and attach harness (take-up) strap assembly (Type II Hood)					
	a. Rivet snap on the left side facing piece as indicated by pattern template with the cap or washer set on the outside.					
	b. Thread loop webbing through dee ring abutting the edges on the underside. Position assembly on the left or right section approximately one inch from back dome seam with the edge of loop end 3/8 (± 1/8) inch above the bottom edge. Stitch the opposite end thru all plies forming a 1-1/4 (± 1/8) inches or two 3/4 (± 1/8) inch combination box-criss-cross stitch.	301	Box-criss-cross stitch	6-8	B or E	B or E
	c. Turn under one end of the long strap and position with loose end towards top of hood on right side facing piece as indicated by pattern template. Stitch thru all plies forming a 1-1/4 (± 1/8) inches or two 3/4 (± 1/8) inch combination box-criss-cross stitch.	301	Box-criss-cross stitch	6-8	B or E	B or E

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TABLE 1 - CONSTRUCTION OF HOODS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
	d. Thread and stitch the free end of the long strap as follows: through dee ring attached to short strap, under and over bar of buckle, through second dee ring, under and over bar of buckle below top ply (as formed). Turn under free end 7/8 to 1 inch and stitch thru all plies forming a 3/4 (± 1/8) inch combination box-criss-cross stitch.	301	Box-criss-cross stitch	6-8	B or E	B or E
	e. The finished length shall be as follows: Strap - 45 (± 2) inches Loop - 2 (± 1/2) inch Loop shall be measured from folded edge of loop end to folded edge of stitched end. Strap shall be measured from fold in webbing at the right side facing piece to free end of strap fully extended.					
12.	Attach snap fasteners to frame Attach the studs and eyelets of snap fasteners to the hood frame to correspond with button caps and sockets on the inner face opening of hood (see Figure 4).					
13.	Assembling a. Assemble helmet in frame and attach frame to face opening of hood.					
	b. Insert the facepiece support and gold facepiece between the inside and outside face opening channel. Position facepiece cover over studs on outer aluminized facepiece and secure snap fasteners (see Figure 3).					
14.	Clean hoods a. Trim all ends of thread and remove all loose thread.					
	b. Remove all spots and stains. NOTE: Care shall be taken in the use of the solvents for cleaning to assure that the coating on the aluminized material is not damaged.					

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3.8 Workmanship. The finished hoods shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article submitted in accordance with 3.2 shall be inspected for compliance with design, construction, workmanship, appearance, and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-1666, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. A certificate of compliance shall be furnished for the following requirements: flame retardant treatment of webbing in 3.3.7.2.2, the alternate webbing, when used, in 3.3.7.2.2, coatings of 3.3.9.2, and frame of 3.3.11.1. In addition, testing shall be performed on the components listed for the characteristics noted in Table II. The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table II shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if one or more sample units or lots average fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

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Lot size	Sample size
800 or less	2
801 thru 22,000	3
22,001 and over	5

The lot size expression and sample units for testing shall be as follows:

Component	Lot Size Expression	Sample Unit
Snaps	Gross	6 snaps
Facepiece support	Each	2 supports
Facepiece, metallized	Each	10 facepieces
Facepiece cover	Each	3 covers

Table II - Testing of components

Component	Characteristic	Requirement	Test Method Paragraph
Snaps	Material identification	3.3.7.2.4	1/
	Style	3.3.7.2.4	Visual
	Dimensions	3.3.7.2.4	Scale
Facepiece support	Vertex and prismatic power	3.3.9.3.1	4.4.1.1
	Optical distortion	3.3.9.3.2	4.4.1.2
Metallized facepiece	Optical distortion	3.3.9.3.2	4.4.1.2
	Luminous transmittance	3.3.9.3.3	4.4.1.3
	Adhesion of coating	3.3.9.3.4	4.4.1.4.1
	Hardness of coating	3.3.9.3.5	4.4.1.4.2
	Infrared transmittance	3.3.9.3.6	4.4.1.5
	Infrared radiant heat resistance	3.3.9.3.7	4.4.1.6
Facepiece cover marking	Resistance to:		
	Water	3.3.9.4.2	4.4.1.7 1/
	Wet and dry crocking	3.3.9.4.2	5651 1/

1/ Unless otherwise specified, a certificate of compliance may be accepted for this characteristic.

4.4.1.1 Vertex and prismatic power. The facepiece support shall be tested for vertex and prismatic power with a telescope lensometer, projection lantern, or any other suitable instrument, to determine conformance to 3.3.9.3.1.

4.4.1.2 Optical distortion. The facepiece support, excluding the area 3/4 inch from the outer edge, shall be viewed from the wearer's position with the unaided eye for visible distortion and optical defects to determine conformance to 3.3.9.3.2. The definition of "unaided eye" shall be a person who possesses 20/20 or better visual acuity with out without spectacles.

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4.4.1.3 Luminous transmittance. The facepiece, excluding the area 3/4 inch from the outer edge, shall be tested in accordance with ASTM-D-1003-61 to determine conformance to 3.3.9.3.3.

4.4.1.4 Durability of coatings. Before and after subjecting the coated facepieces to any durability of coating test, the coated facepiece shall be thoroughly and carefully cleaned to remove dirt, film, fingermarks, and grease using mild soapy water followed by a thorough rinsing and drying with a soft, lintless, facial tissue pad.

4.4.1.4.1 Adhesion of coating. A strip of cellophane tape shall be firmly pressed to the center of the coated surface parallel to the 8-1/4 inch side starting at one edge and allowing 1-3/4 inches of the tape to extend over the other edge. The cellophane tape shall be 1 inch in width by 10 inches in length and shall possess a minimum adhesion value of not less than 40 ounces per inch of width. The free end of the tape shall be double backed at 180 degrees and pulled parallel to the facepiece surface. The tape shall be removed quickly with a snap action which exerts the greatest possible stripping effect on the coatings. The tested facepiece area shall be visually examined for conformance to 3.3.9.3.4. Examination of the cellophane tape surface will aid in determining whether any of the coating has been removed.

* 4.4.1.4.2 Hardness of coatings. A pad of bleached cheesecloth (previously laundered), one inch in diameter, 1/2 inch in thickness, shall be rubbed for 25 cycles lengthwise across the gold coated surface of a 3-1/2 by 8 inch specimen. The 8 inch side shall be considered the lengthwise direction for this test. Either a stationary or dynamic force of two pounds shall be applied. If a stationary pressure is chosen, then the gold coated surface of the specimen shall be cycled in its lengthwise direction under the stationary abradant (cheesecloth pad) for 25 cycles at one second per cycle. If a dynamic pressure is chosen, the abradant (cheesecloth pad) shall be rubbed lengthwise across the gold coated surface of the specimen for 25 cycles at one second per cycle. A cycle shall consist of a rubbing motion extending from one edge to the extreme opposite edge and back. The area shall be examined for conformance to 3.3.9.3.5.

4.4.1.5 Infrared transmittance. The facepiece shall be tested for infrared transmittance with a double beam infrared spectrophotometer capable of providing a continous reading of 1 - 7.5 microns, or as an option, from 1 - 3 microns, the readings shall be taken at 0.1 micron intervals and for 3 - 7.5 microns, the readings shall be taken at 0.5 micron intervals to determine conformance to 3.3.9.3.6. A 2 by 8 inch strip is cut from each facepiece. From the latter, four 2 by 2 inch squares are cut. Two of the squares are treated with aqual regia to remove gold coating. The plastic substrate is immediately washed with distilled water and dried. In cases where a plastic overcoat is present, more time will be required before the gold is dissolved and the overcoat will begin to separate. The plastic substrates must be carefully handled because they will be employed in the adjustment of the instrument for compensation. Positioning the clear plastic substrate in the instrument (one in each channel) the wavelength is set at a particular value of interest. All instrumental settings are made for optimum response between zero and 100%. The gold-coated sample is then inserted in place of the substrate in the sample beam and its transmittance recorded for the particular wavelength setting. In all measurements, the substrate in the reference beam is never removed.

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4.4.1.6 Infrared radiant heat resistance. The apparatus for conducting the heat resistance tests shall be in accordance with Figure 1 of MIL-C-87076. The heat source shall consist of a bank of five, 500 W, infrared, tubular, translucent quartz lamps having a 5 inch lighted length and a mean overall length of 8-13/16 inches. The lamps shall be mounted so that the lamp's surfaces are approximately 0.015 inch apart. The bank or array shall be mounted and centered behind a 2-1/4 by 5-1/2 inch cutout on 1/2 inch transite board. A specimen holder plate with a 2-1/2 by 6 inch center cutout shall be positioned so that the distance from the nearest lamp surfaces to the test specimen is exactly one inch. The quartz lamp shall be heated electrically and the power input controlled by means of a variac having a capacity of at least 25 amps. A voltmeter, accurate to $\pm$ 1 volt shall be installed on the load circuit to indicate operating or load voltage to the lamps. The quartz lamp shall be operated at 82 operating volts and the temperature indication of the array shall be 2745°F $\pm$ 10°F (1507 $\pm$ 6°C). The individual lamps in the array shall not vary more than $\pm$ 25°F ($\pm$ 14°C) from the average temperature. Adjustment may be made in the variac setting to produce the required temperature. Temperature indications shall be made at the center of the lamps using an optical type pyrometer.

Test procedure. The test specimen shall be approximately 3-1/2 by 8 inches. Before positioning specimen holder in front of the heat source, apply current to the lamps and adjust variac to obtain 82 operating volts as indicated by the voltmeter. The individual lamp temperature shall be determined by a properly scaled optical type pyrometer. The temperature indications shall be taken along the center line of the individual lamps, not between the lamps. The average temperature of the array shall be 2745° $\pm$ 10°F (1507 $\pm$ 6°C) and the individual lamps shall not vary more than $\pm$ 25°F ($\pm$ 14°C) from the average temperature. Adjustments may be made in the variac setting to obtain the specified temperature. The variac setting shall be undisturbed and current to the lamps shall be shut off. The side of the metallized facepiece marked "outside" shall be considered the face for test purposes. The specimens shall be placed on the back of the face plate of the specimen holder so that the face of the test specimen completely covers the window of the holder plate and faces the heat source when positioned in front of the quartz lamp array. The test specimen shall be held securely to the holder plate by taping all four sides with a pressure sensitive tape. The tape should not overlap the window or be exposed directly to radiation. The distance from the specimen to the nearest edges of the lamp surface shall be exactly one inch. Current shall be applied to the lamps and the specimen shall be exposed for 300 seconds. Timing shall start when the current is applied to the lamps and terminate when the current is shut off. The specimen holder shall be removed from the apparatus and the test area visually examined for conformance to 3.3.9.3.7.

Report. Five specimens shall be tested from each unit of product. Failure of any one specimen shall constitute failure of the unit of product.

4.4.1.7 Water resistance. One hour after marking, place facepiece cover with printed instructions in distilled water at 70°F (21.1°C) for 4 hours, Remove from the water and air dry for one hour. The facpiece cover marking shall be visually examined for conformance to 3.3.9.4.2.

4.4.2 Examination of the end item. The end item examination provisions for the finished hoods shall be as specified in MIL-STD-1666.

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4.4.3 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

5. Packaging

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each hood, with the bottom neatly folded inside the dome of the hood, shall be individually packaged in a snug-fitting clear polyethylene film bag having a film thickness of not less than 0.003 inch. All seams and closures shall be formed by heat sealing. A 1/4 inch diameter hole shall be made at one corner of each bag to allow excess air to escape. As an alternate, the polyethylene bag may be of the tuck-in or reverse flap type, in which a heat seal closure and corner hole are not required. The open end of the bag shall be of sufficient length so that it can be folded over. The bag may be fabricated from polyethylene film tubing or sheeting. Each hood shall then be placed in a snug-fitting fiberboard box conforming to style RSC, type CF, class domestic, variety SW, grade 200 of PPP-B-636. The top and bottom flaps of the box shall be securely sealed with gummed paper tape, 3 inches in width conforming to type III, grade B of PPP-T-45.

5.1.2 Level C (Commercial packaging). Covers and hoods shall be packaged to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

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5.2.1 Level A. Six (6) hoods, of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container, as follows: Place the packages flat, three along the length, one across the width and two in depth, with the length of the packages parallel to the end panels of the container. The shipping container shall conform to style RSC, type SF, class weather-resistant, grade V2s of PPP-B-636. The shipping container shall be closed, waterproofed and reinforced in accordance with method V requirements of the appendix of the container specification.

5.2.2 Level B. Six (6) hoods, of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container, as follows: Place the packages flat, three along the length, one across the width, and two in depth, with the length of the packages parallel to the end panels of the container. The shipping container shall conform to style RSC, type CF, class domestic, variety SW, grade 275 of PPP-B-636. The shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification.

5.2.3 Level C (Commercial packing). Covers and hoods, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Palletization. When specified (see 6.2), hoods packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means K and L or bonding means P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The hoods are intended to be worn as follows:

Type I hood is to be used with Coat, Trousers, and Gloves, Aluminized, Proximity as part of an ensemble for personnel rescue and fighting aircraft fires on air fields.

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Type I hood is to used with the Coveralls and Gloves, Aluminized, Proximity as part of an ensemble for personnel rescue and fighting aircraft fires aboard aircraft carriers.

NOTE: The cover is intended as a replaceable component to the Type I hood only (see 3.3.10).

Type II hood (used to protect the Oxygen Breathing Apparatus) is to be used with Coveralls and Gloves, Aluminized, Proximity as part of an ensemble for damage control aboard ship.

The three ensembles rely on the simultaneous use of water and foam and any other firefighting or damage control equipment to provide maximum protection while the wearer approaches and operates close to the fire.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Type of hood required (see 1.2)
c. When cover, required for Type I hood (see 3.3.10), is required
d. When first article is required (see 3.2), the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity and testing and approval of the first article.
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2)
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1)
g. When palletization is required (see 5.3)

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

6.4 Flame resistant treatments. Only those flame resistant treatments, approved by the U.S. Army Natick Research and Development Center, Natick, MA 01760, and the appropriate medical service, and so listed in the invitation for bids or request for proposal, shall be considered acceptable for the related procurement.

6.5 Equal item. Prior to the use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible Military Agency.

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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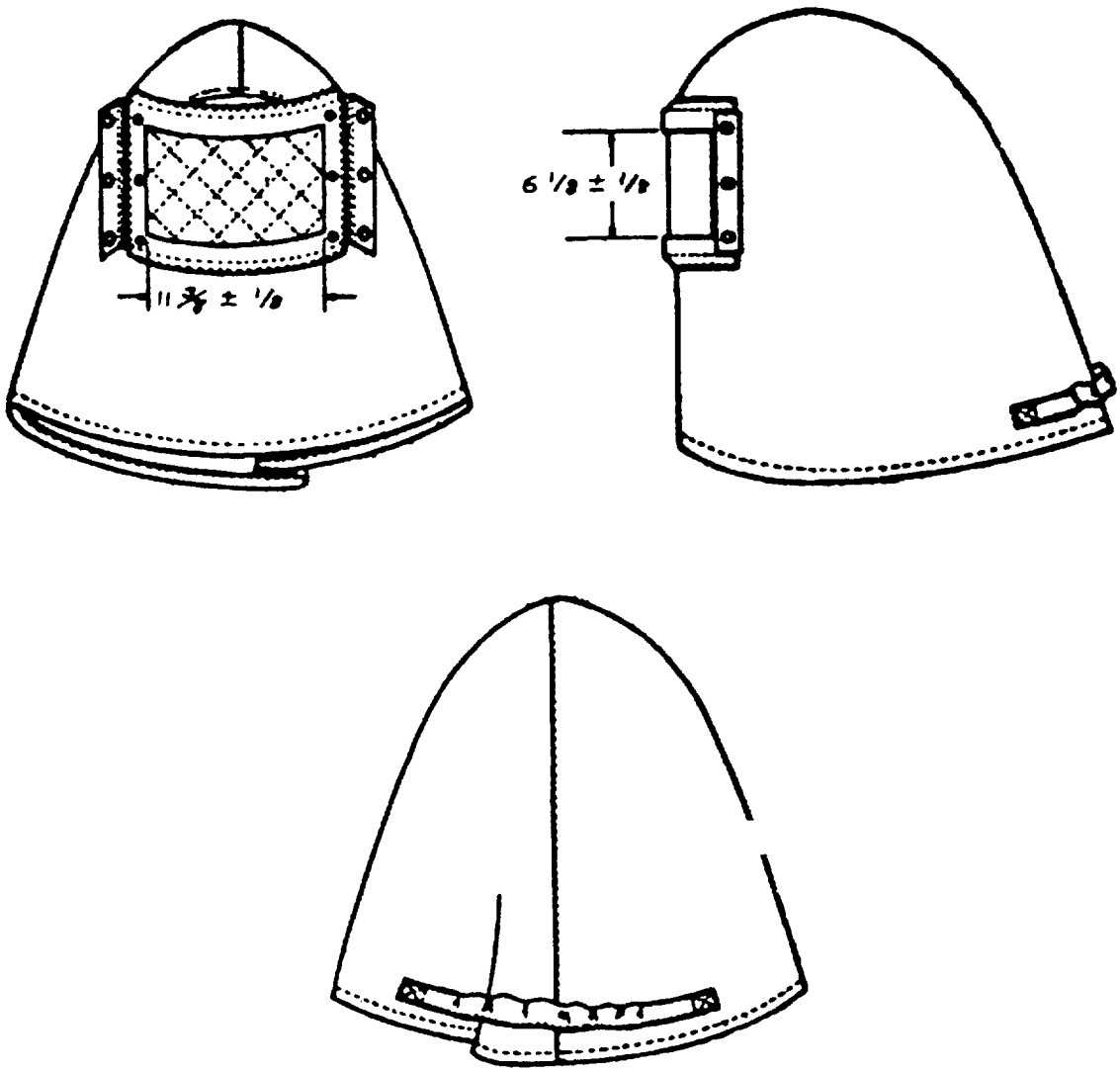
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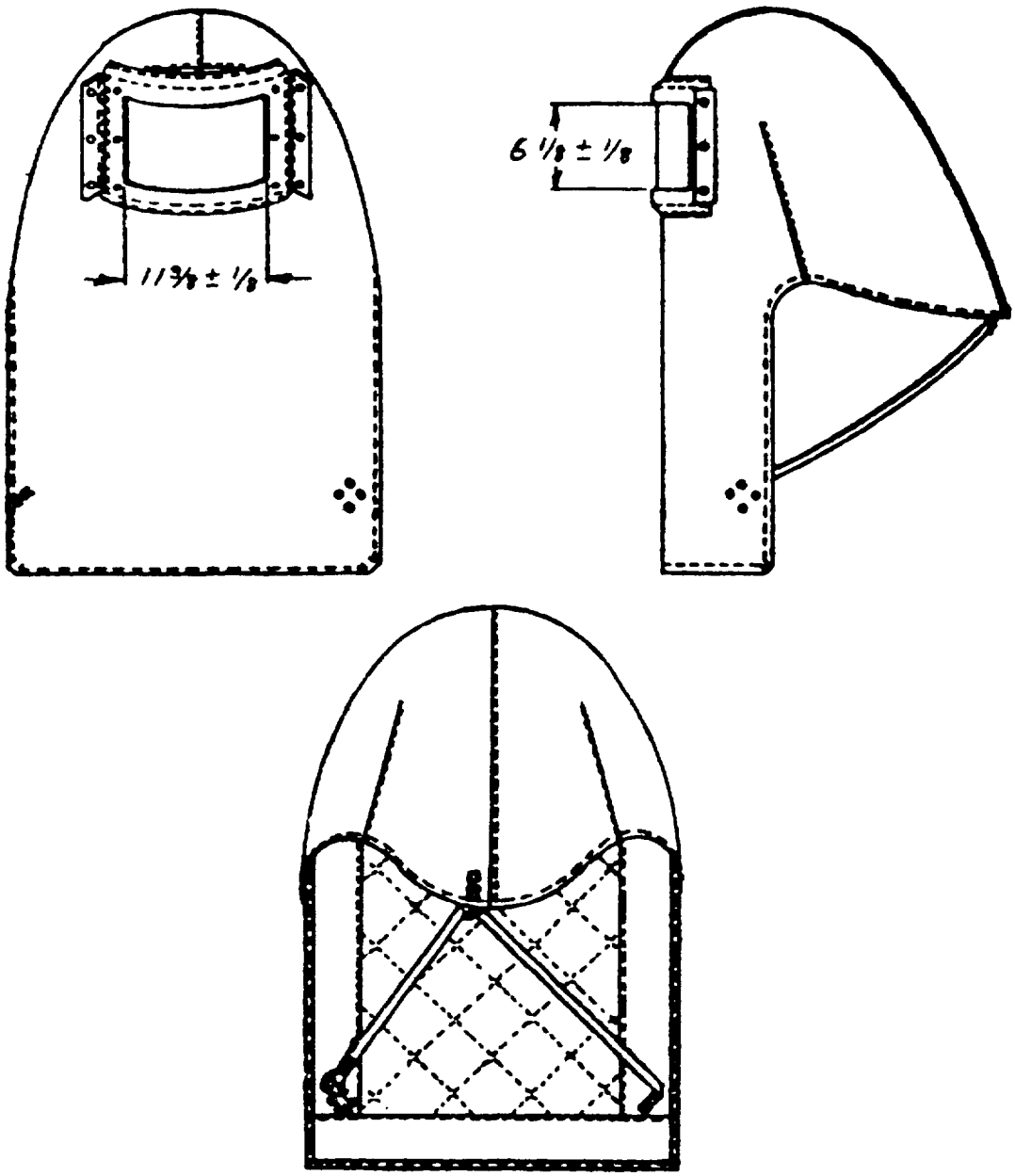
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FIGURE 1. HOOD, FIREMAN'S, ALUMINIZED, PROXIMITY (TYPE I - SHOULDER LENGTH - AIRCRAFT FIREFIGHTING)

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FIGURE 2. HOOD, FIREMAN'S, ALUMINIZED, PROXIMITY
(TYPE II - WITH FRONT APRON - SHIPBOARD DAMAGE CONTROL)

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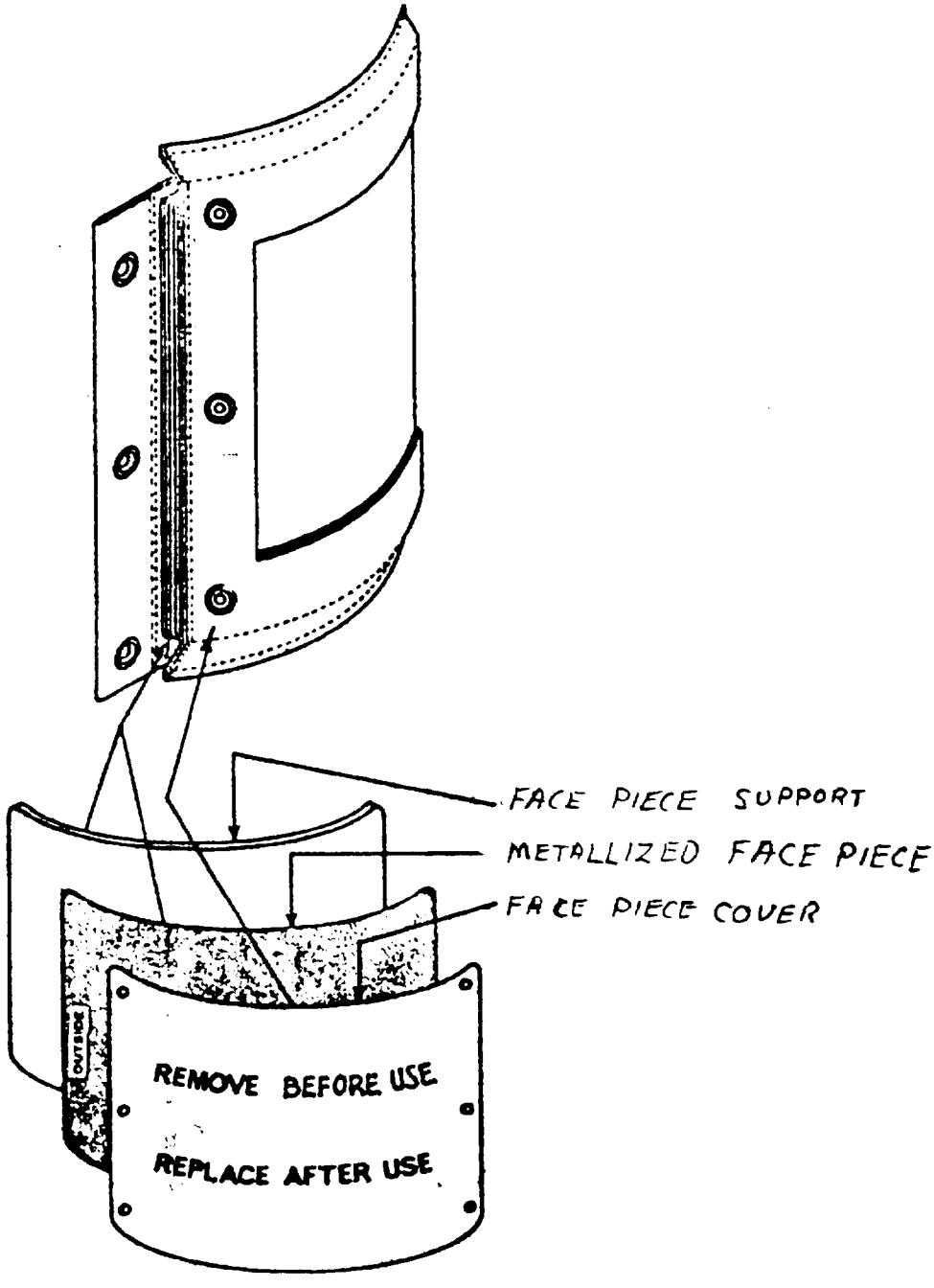


FIGURE 3 - FACE PIECE OPENING INSERTS

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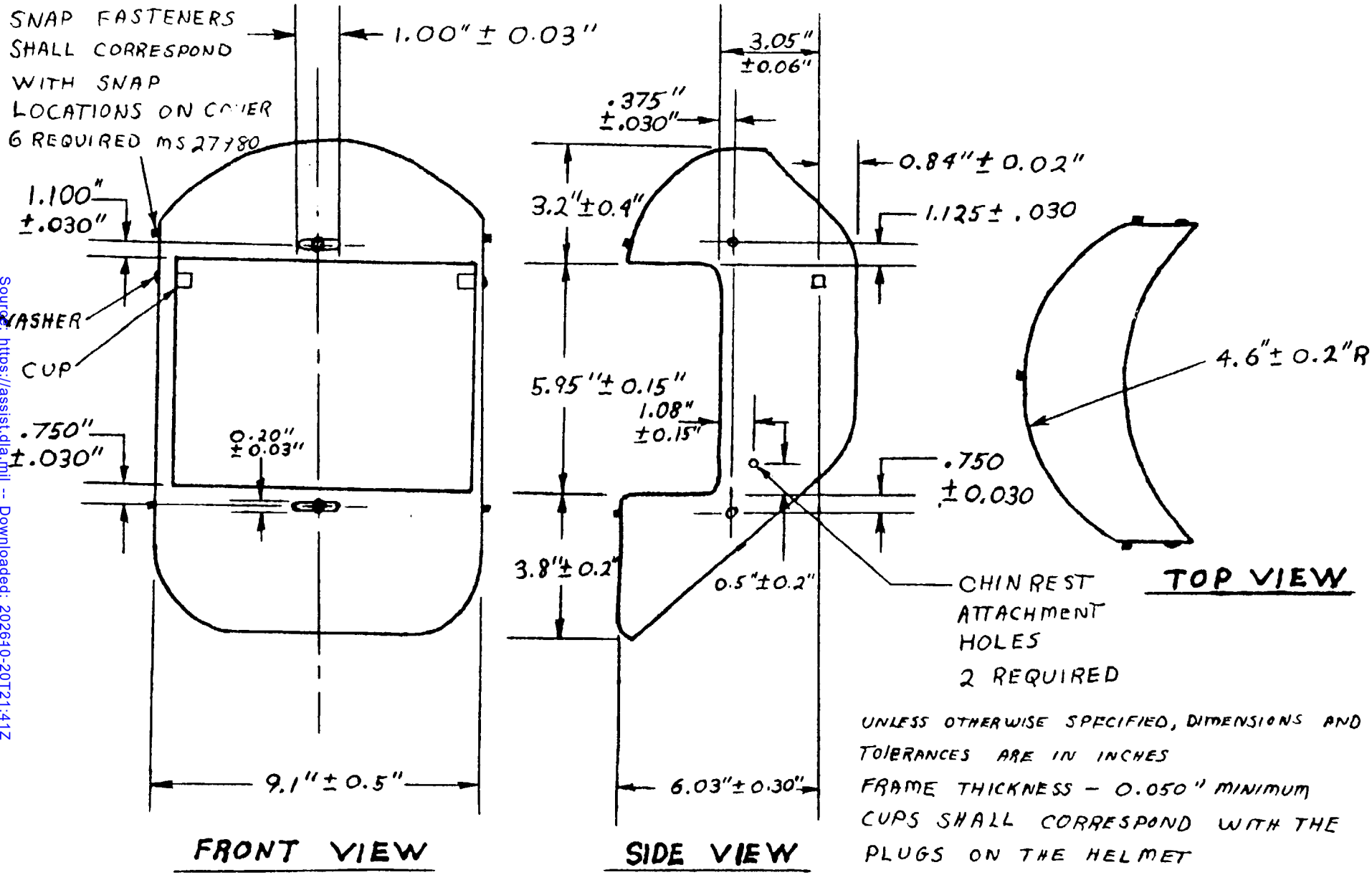
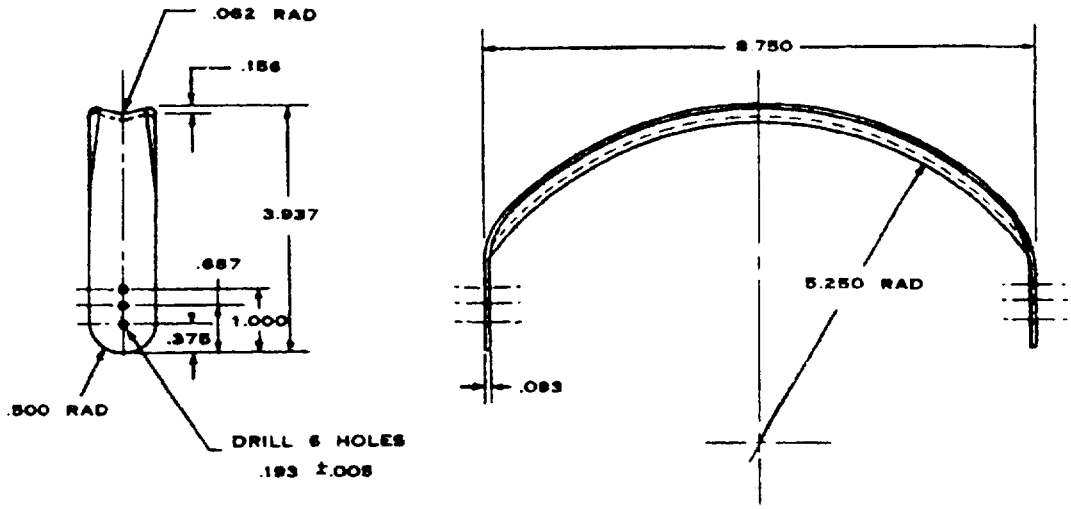


FIGURE 4 – FRAME

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FIGURE 5 - CHIN REST

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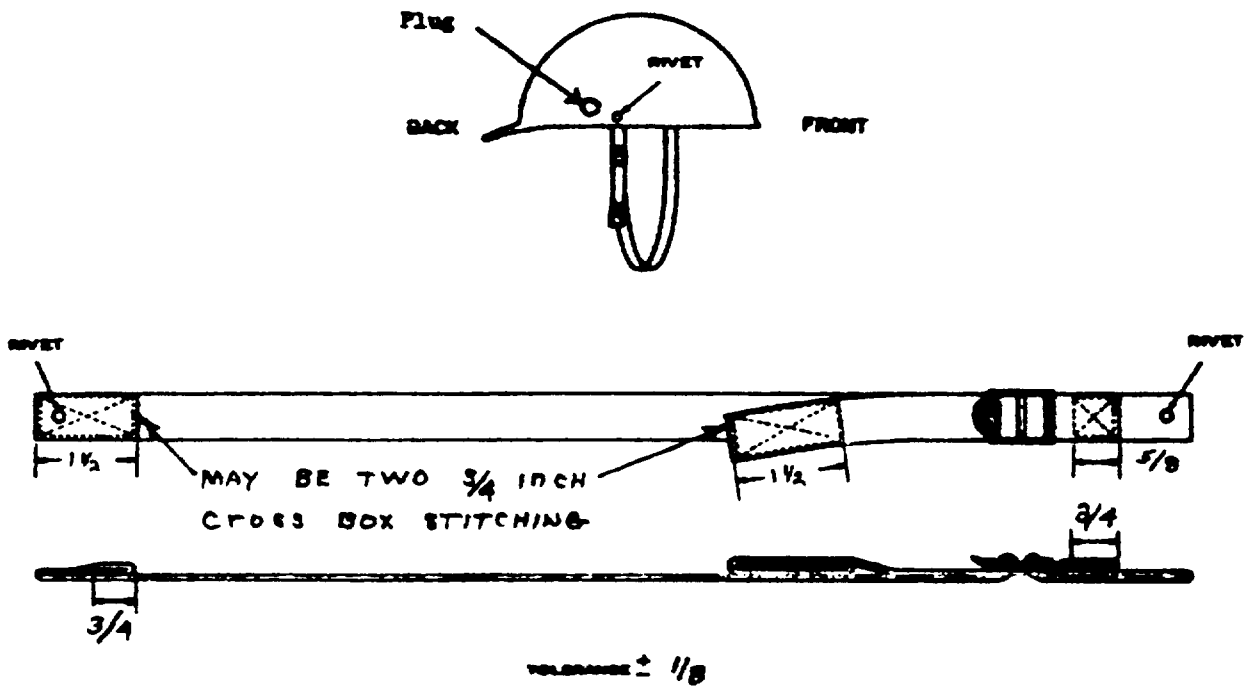
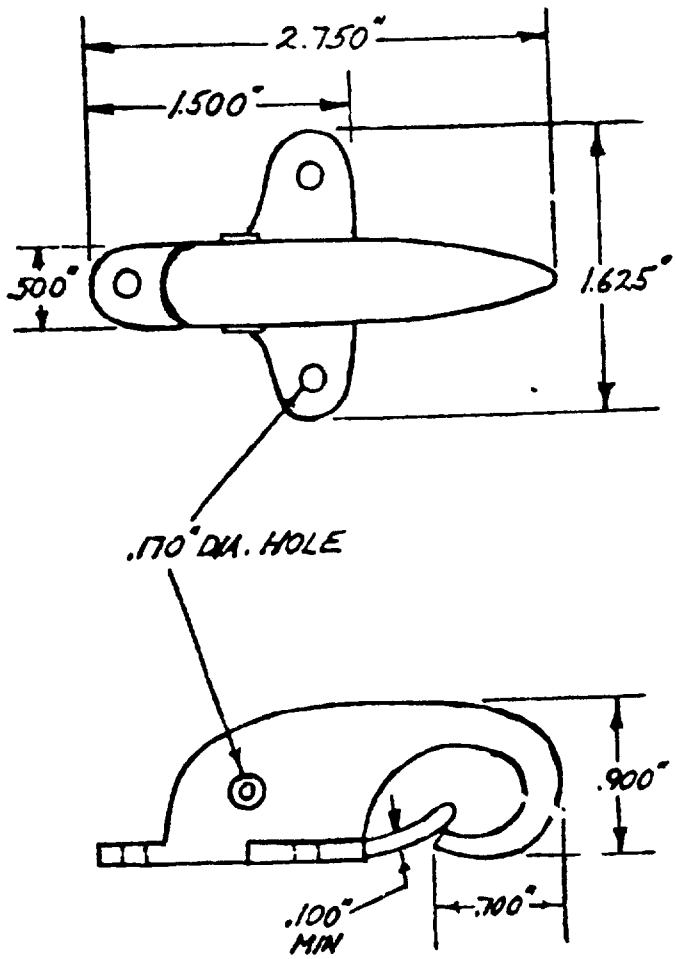


FIGURE 6 - CHIN (TAKE-UP) ADJUSTMENT STRAP ASSEMBLY

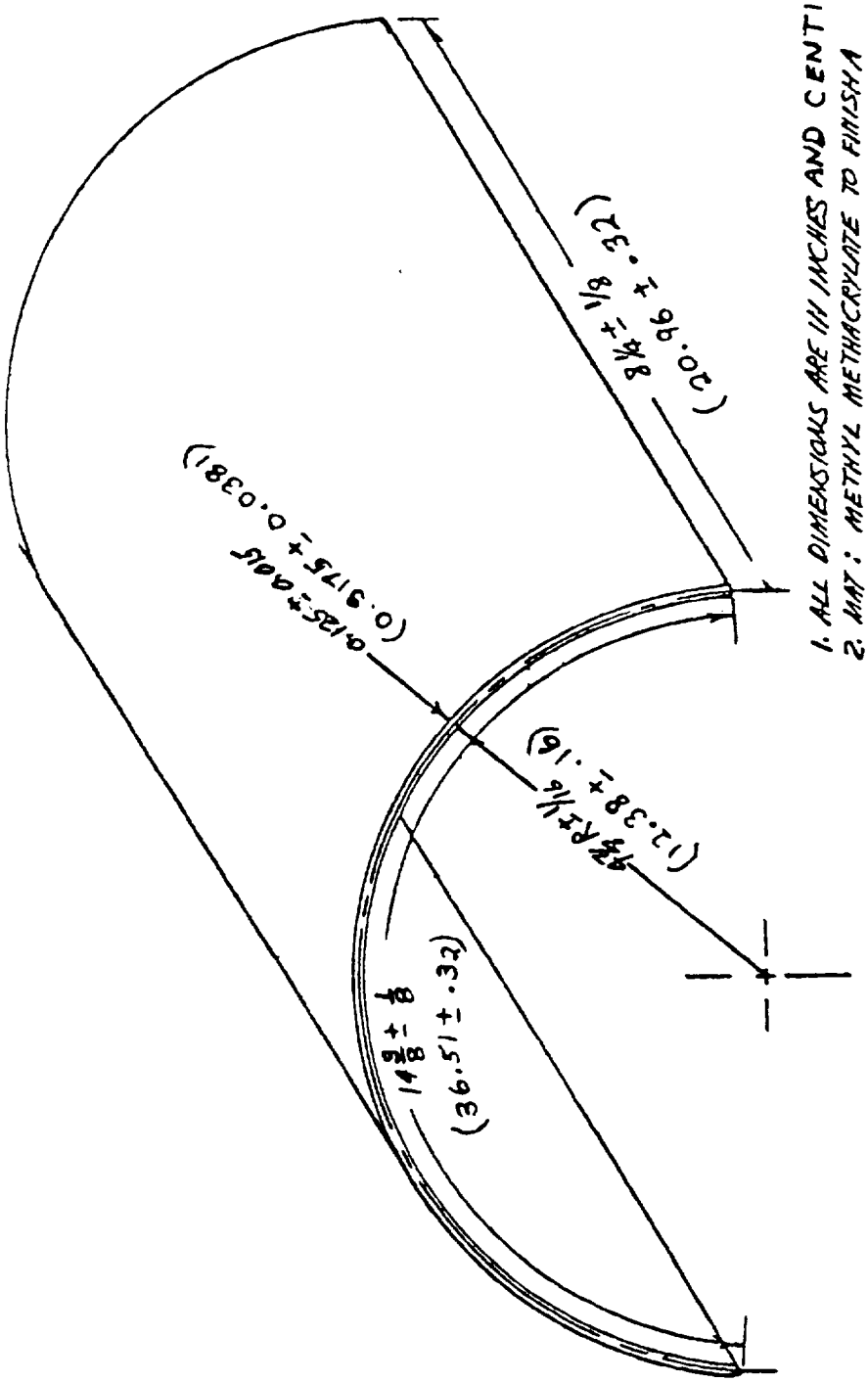
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TOLERANCE ± .010 INCH

FIG. 7 SNAP

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1. ALL DIMENSIONS ARE IN INCHES AND CENTIMETERS
2. MAT'L: METHYL METHACRYLATE TO FINISH A OF MIL-P-8184
3. ALL EDGES SHALL BE SMOOTHED AND ROUNDED
4. THE FACEPIECE SUPPORT SHALL BE THERMOFORMED FROM SHEET STOCK AND ANNEALED

FIG. 8 - FACEPIECE SUPPORT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-29144B	HOOD, FIREMEN'S, ALUMINIZED, PROXIMITY

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS (Street, City, State, ZIP Code)

4. TYPE OF ORGANIZATION (Mark one)

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER (Specify): _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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